SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Global Dark and Light Mode Theme Implementation

Student List

Field	Details
Change Request ID	CR-012
Change Request Title	Global Dark and Light Mode Theme Implementation
Student Name	Y.Ugatharsini
Student ID	JF/ICT/24/33
Implementation Status	Completed
Date	10 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System previously used a fixed visual theme across its user interface. Users needed the ability to switch between light mode and dark mode to improve readability, accessibility, and user comfort under different lighting conditions.

The change was required to provide a consistent theme across all major pages of the application and to allow users to retain their preferred theme when navigating between pages or refreshing the browser.

1.2 Existing Behaviour

Before implementation:

- The application primarily used fixed light-theme colors.
- Backgrounds, text, cards, forms, and tables used predefined CSS colors.
- No global theme-switching facility was available.
- Some page elements could not adapt automatically to a different theme.
- Theme preferences were not preserved between page refreshes.
- Users could experience poor text visibility when backgrounds and text colors did not adapt together.

1.3 Requested Change

A global dark/light mode feature was required for the Student Management System.

The system must:

- Provide a control to switch between light mode and dark mode.

- Apply the selected theme across Home, Login, Admin Dashboard, and Student Dashboard pages.
- Change page backgrounds, text, cards, forms, tables, borders, and other relevant interface elements.
- Maintain consistent theme settings during navigation.
- Save the user's theme preference using browser local storage.
- Restore the saved preference after refreshing the page.
- Ensure text remains readable in both themes.
- Preserve existing authentication, student management, and CRUD functionality.
- Avoid unnecessary backend or database changes.

2. Implementation

2.1 Implementation Summary

CR-012 was implemented as a global theme-management enhancement using React Context, React state, browser local storage, and CSS custom properties.

A shared theme context was introduced to manage the current theme. The selected theme is applied to the document root using the `data-theme` attribute.

CSS variables are used to control background colors, text colors, borders, input colors, table headers, hover states, and shadows. This allows the application interface to adapt to the selected theme.

A theme-toggle control was added to allow users to switch between light mode and dark mode. The selected preference is saved in local storage so that it can be restored when the application is refreshed.

Additional CSS adjustments were made to improve text visibility and maintain readable contrast in dark mode, particularly in the administrator dashboard, forms, and student table.

2.2 Affected Files

File	Change	Purpose

frontend/src/App.jsx	Modified	Wraps the application routes with the shared ThemeProvider.
frontend/src/context/ThemeContext.jsx	Added / Modified	Manages the global theme state and saves the selected preference.
frontend/src/components/ThemeToggle.jsx	Added / Modified	Provides the control for switching between light and dark modes.
frontend/src/components/Navbar.jsx	Modified	Integrates the theme-toggle control into the navigation interface.
frontend/src/styles/index.css	Modified / Verified	Defines global theme variables, transitions, and theme-specific styles.
frontend/src/styles/dashboard.css	Modified	Applies theme-aware colors to the administrator dashboard, forms, tables, and text.
frontend/src/styles/login.css	Modified / Verified	Supports theme-aware login backgrounds, cards, labels, and inputs.
frontend/src/styles/landing.css	Modified / Verified	Supports theme-aware Home page backgrounds and content.
frontend/src/styles/studentdashboard.css	Modified / Verified	Supports theme-aware student dashboard components.

2.3 Technical Changes

2.3.1 Global Theme Management

A shared React Context was used to manage the theme across the application. This allows different pages and components to use the same theme state instead of maintaining separate theme settings.

The ThemeProvider provides the current theme and a function to toggle between light and dark modes. Application routes are rendered inside the provider to maintain shared theme state.

2.3.2 Theme Toggle

A theme-toggle button was introduced to switch between light mode and dark mode.

The control updates the current theme and applies the corresponding styles to the application.

The theme-toggle control is designed to make theme switching accessible and straightforward for users.

2.3.3 CSS Custom Properties

CSS custom properties were implemented in `global.css` to define reusable theme colors for the Dark Mode and Light Mode interfaces.

The principal variables include:

- --bg-primary
- --bg-secondary
- --bg-tertiary
- --text-primary
- --text-secondary
- --border-color
- --input-bg
- --input-text
- --primary-color
- --primary-hover
- --table-header-bg
- --hover-bg
- --shadow-color

These variables allow interface colors to change consistently when the selected theme changes. The theme settings are managed through ThemeContext.jsx and applied across the interface using the relevant CSS files.

2.3.4 Theme Persistence

The selected theme is stored in browser local storage using the key sms-theme.

When the application starts, the saved preference is read and applied. This allows the user's selected theme to remain active after refreshing the page.

If local storage is unavailable, the theme initialization logic falls back to the default light theme.

2.3.5 Text Visibility and Readability

Theme-specific CSS rules were added to ensure that text remains visible against the selected background.

Dashboard headings, paragraphs, table content, labels, and supporting text use theme-aware colors. Form fields and other interface elements are also styled to maintain readability in dark mode.

This prevents dark text from remaining difficult to read against dark backgrounds.

2.3.6 Cross-Page Consistency

The shared theme is designed to apply consistently across the following pages:

- Home page
- Login page
- Administrator dashboard
- Student dashboard

Switching the theme updates the shared theme state rather than changing only an individual page. The same preference remains active when users navigate between pages.

2.4 Implementation Logic

The implementation follows these steps:

1. Initialize the global theme using the saved preference in local storage.
2. Provide the theme state through the React Context provider.
3. Render the application routes inside the shared provider.
4. Allow the user to switch between light mode and dark mode.
5. Update the shared theme state when the toggle is activated.
6. Apply the selected theme using the document root's data-theme attribute.
7. Use CSS custom properties to update backgrounds, text, cards, tables, forms, and borders.
8. Save the selected theme in local storage.
9. Restore the saved theme after a page refresh.
10. Verify that the theme is consistent across all major application pages.
11. Verify that text remains readable in both light and dark modes.
12. Confirm that existing authentication and student management functionality remains unchanged.

2.5 Error and Data Handling

The theme feature is implemented on the frontend and does not require changes to the application's database or backend APIs.

The theme initialization logic handles local storage access safely. If local storage is unavailable, the application can fall back to the default light theme.

The feature does not modify student records, login credentials, user roles, authentication logic, or existing CRUD operations.

The existing application routes and backend operations remain separate from the thememanagement functionality.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend testing.

The testing approach focused on checking the theme-toggle functionality, theme persistence, cross-page consistency, text readability, and preservation of existing application functionality.

Testing Types:

- Functional testing
- UI testing
- Theme consistency testing
- Persistence testing
- Readability testing
- Navigation testing
- Regression testing

Environment: Development

Browser: [Chrome / Edge / Firefox]

Test Roles: Guest, Administrator, Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the application for the first time.	The default theme is displayed correctly.	Default theme displayed correctly.	Pass
TC-02	Switch from light mode to dark mode.	The interface changes to dark mode.	Dark mode applied correctly.	Pass
TC-03	Switch from dark mode to light mode.	The interface changes to light mode.	Light mode applied correctly.	Pass
TC-04	Change theme on the Home page.	Theme-aware Home page elements update correctly.	Home page theme changed correctly.	Pass
TC-05	Open the Login page after changing the theme.	The Login page uses the selected theme.	Login page theme applied correctly.	Pass

TC-06	Open the Admin Dashboard.	Dashboard backgrounds, text, forms, and tables use the selected theme.	Dashboard elements displayed correctly.	Pass
TC-07	Open the Student Dashboard.	Student dashboard elements use the selected theme.	Student dashboard theme applied correctly.	Pass
TC-08	Refresh the browser.	The previously selected theme is restored.	Saved theme restored correctly.	Pass
TC-09	Navigate between application pages.	Theme settings remain consistent across pages.	Theme remained consistent.	Pass
TC-10	Inspect text and form-field visibility in dark mode.	Text, inputs, and labels remain readable.	Text and form fields were readable.	Pass
TC-11	Perform existing login and logout operations.	Authentication functionality continues to work.	Authentication functionality remained operational.	Pass
TC-12	Perform existing student CRUD operations.	Create, Read, Update, and Delete functionality remains unchanged.	Student CRUD functionality remained operational.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12

Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Users can switch between light mode and dark mode.	Theme toggle test	Met
AC-002	The selected theme applies across all major application pages.	Cross-page verification	Met
AC-003	Backgrounds, text, cards, forms, and tables adapt to the selected theme.	UI inspection	Met
AC-004	Theme preference persists after refreshing the application.	Local storage and refresh test	Met
AC-005	Theme settings remain consistent during navigation.	Navigation test	Met
AC-006	Text and form controls remain readable in dark mode.	Readability testing	Met
AC-007	Existing authentication functionality remains unchanged.	Regression testing	Met
AC-008	Existing student CRUD functionality remains unchanged.	CRUD regression testing	Met
AC-009	The feature does not require backend or database changes.	Code review	Met

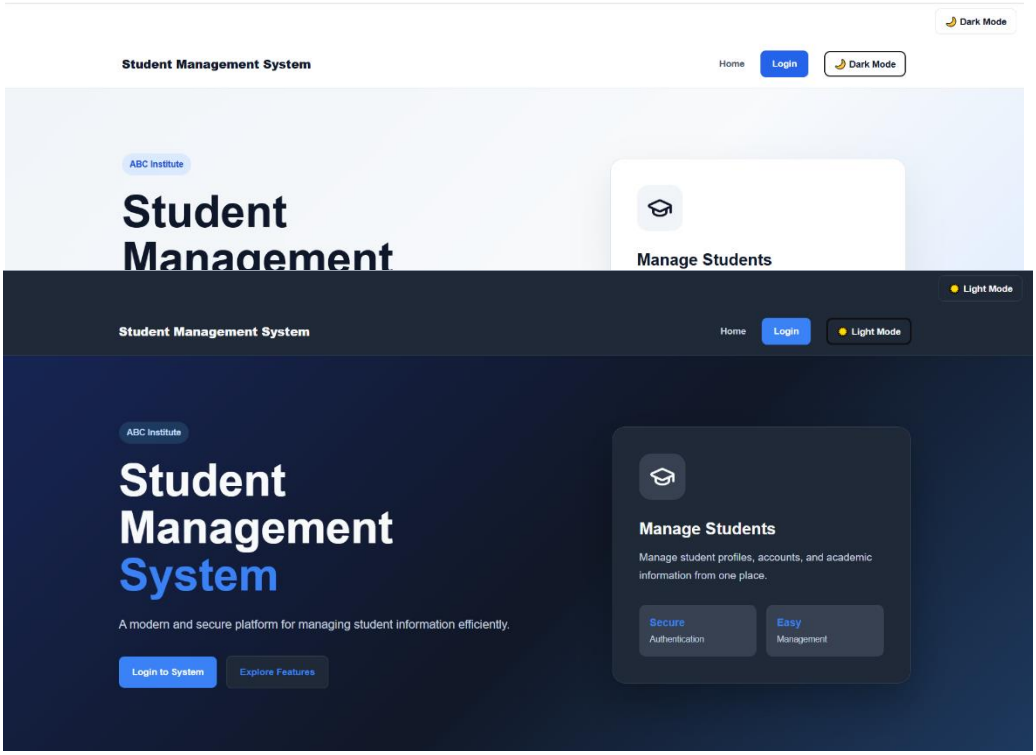
5. Evidence

Evidence 01 – Light Mode and Dark Mode

Related Acceptance Criteria: AC-001, AC-003, AC-006

Description: Screenshot evidence demonstrating the Student Management System in both light mode and dark mode, including readable text and correctly styled interface elements.

Figure 1: Student Management System displaying light mode and dark mode.

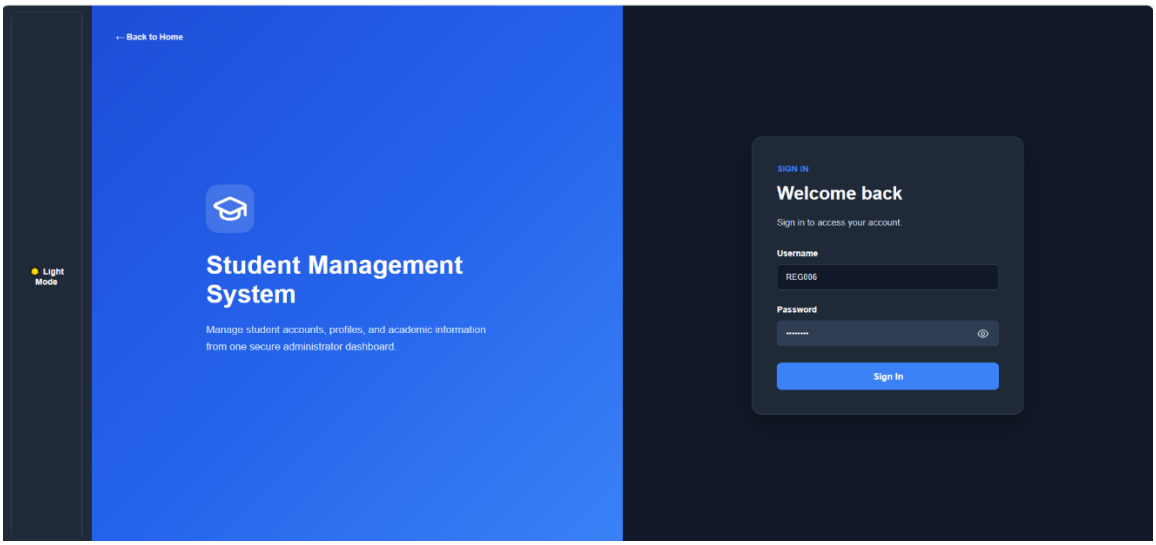


Evidence 02 Cross-Page Theme Consistency

Related Acceptance Criteria: AC-002, AC-003

Description: Screenshot evidence demonstrating that the selected theme is consistently applied across the major application pages, including the Home page, Login page, Admin Dashboard, and Student Dashboard.

Figure 2: Global theme consistency across application pages.



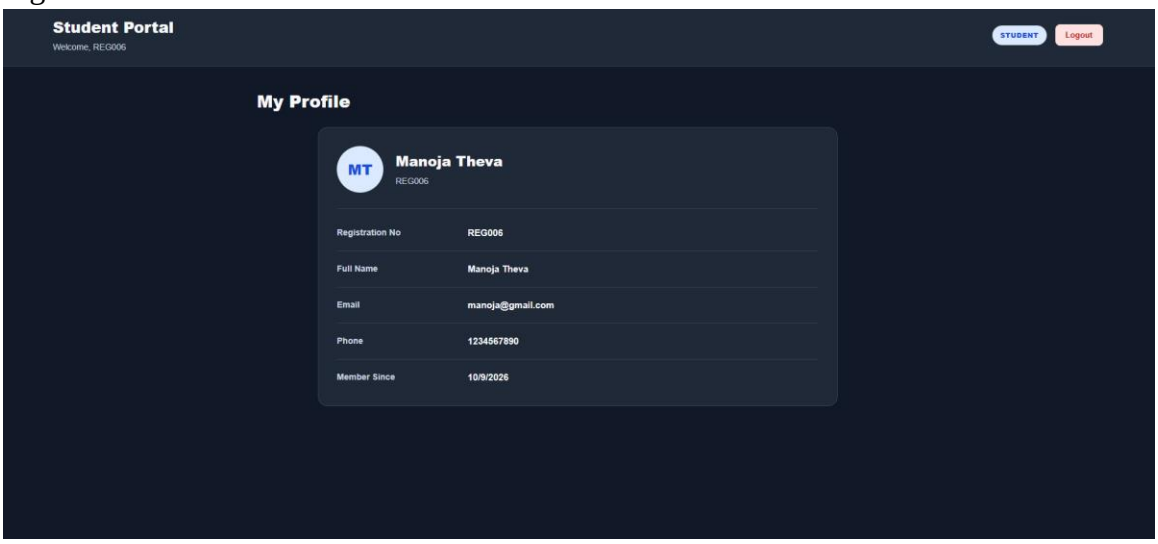
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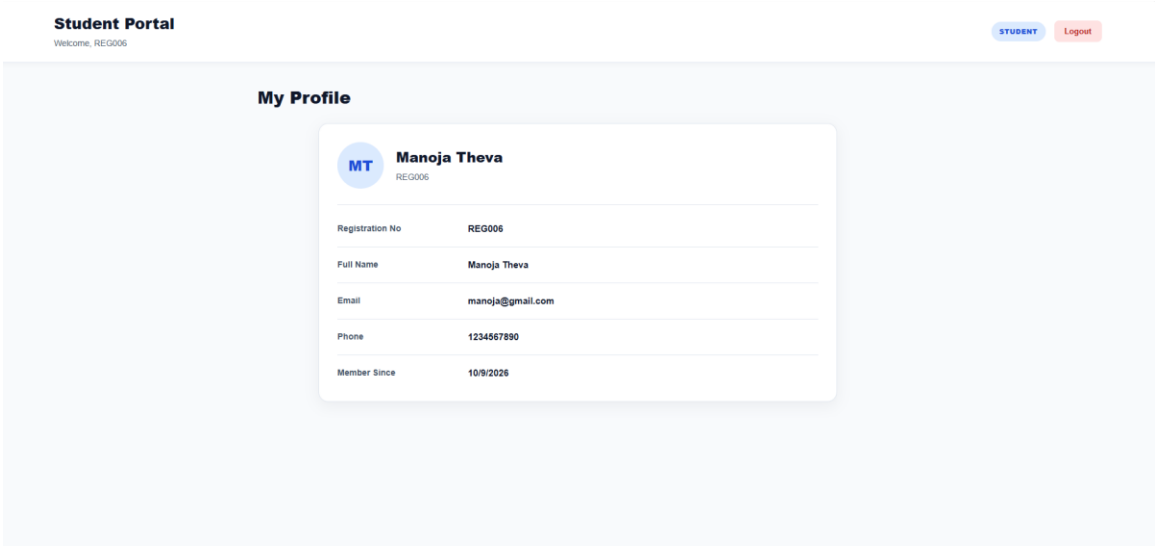
Evidence 03 Theme Persistence After Browser Refresh

Related Acceptance Criteria: AC-004

Description: Screenshot evidence demonstrating that the application restores the previously selected theme after the browser is refreshed.

Figure 3: Selected theme restored after browser refresh.





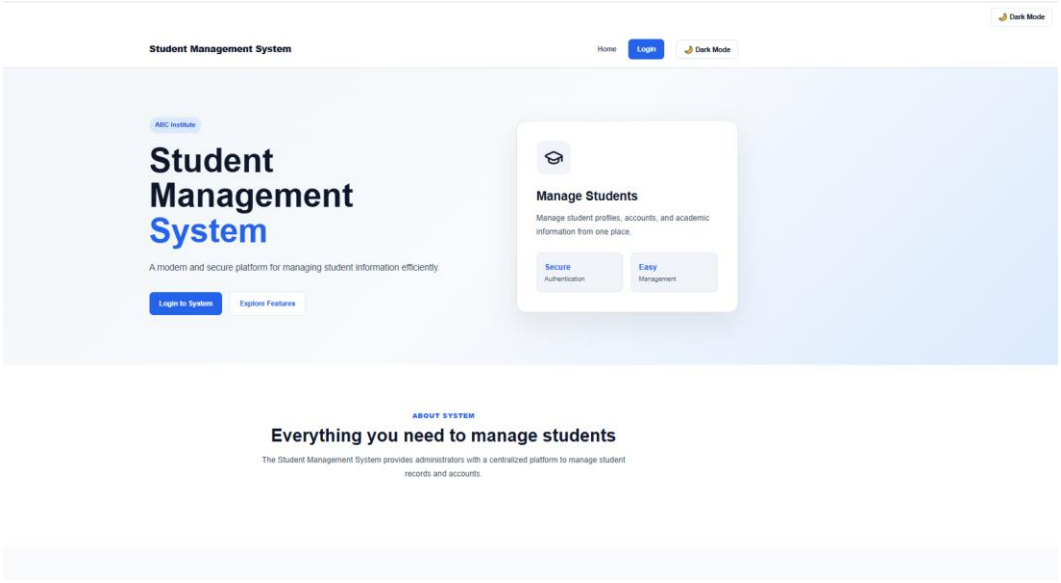
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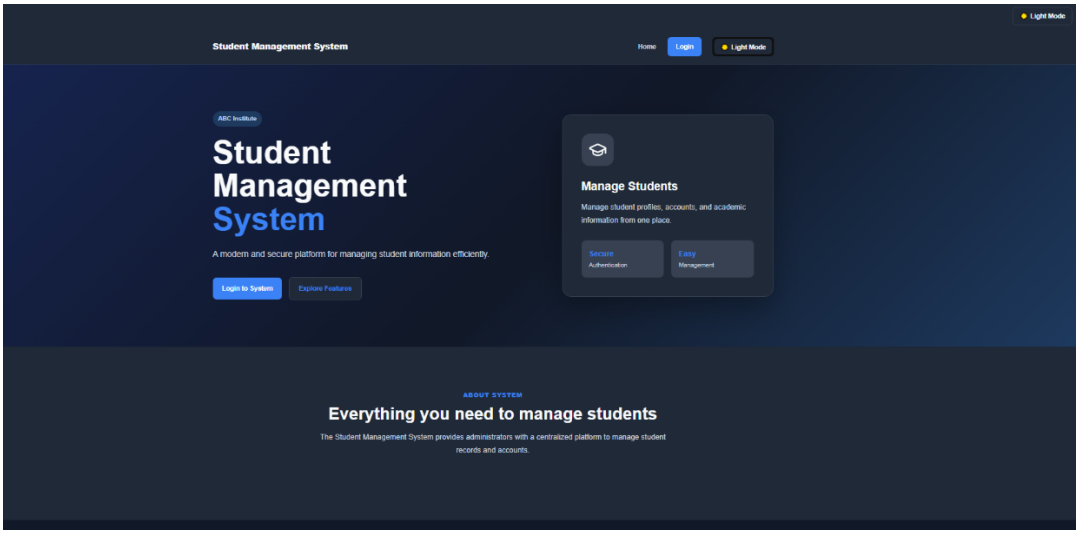
Evidence 04 Navigation Between Application Pages

Related Acceptance Criteria: AC-002, AC-005

Description: Screenshot evidence demonstrating that the selected theme remains consistent when navigating between application pages without requiring the user to select the theme again.

Figure 4: Theme consistency during application navigation.





6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Theme Persistence	Verified
Existing Functionality	Verified
Final Status	Accepted

6.1 Final Summary

CR-012 was successfully implemented to introduce a global dark and light mode feature for the Student Management System.

The implementation uses shared React Context state, CSS custom properties, and browser local storage to manage the selected theme. A theme-toggle control allows users to switch between light mode and dark mode, while the shared theme state maintains consistency throughout the application.

The selected theme applies across the Home page, Login page, Administrator Dashboard, and Student Dashboard. Theme-aware styling supports backgrounds, text, cards, forms, tables, borders, and other interface elements. Additional text visibility adjustments help maintain readability in dark mode.

The selected theme persists after a page refresh and remains consistent during navigation. Existing authentication, backend APIs, database operations, and student CRUD functionality remain unchanged.

All 12 test cases passed, with zero failures and a 100% pass rate. All nine acceptance criteria were met. Therefore, the implementation is considered complete, and the change request is accepted.

6.2 Issues / Limitations

No significant issues identified.

The theme preference is stored in browser local storage, so it is specific to the current browser and device rather than being synchronized through the user's account.

Student Name: Y.Ugatharsini

Student ID: JF/ICT/24/33

Date: 10 October 2026

Signature: Y.Ugatharsini

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